

Logan McDougall

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Summary

Seasoned and performance-focused Software Engineer with over a decade of experience specializing in backend engineering, distributed systems, and infrastructure optimization across enterprise and high-availability systems. Deeply proficient in C# and Java, with extensive experience deploying microservices and modern APIs on AWS and Azure. Adept at scaling legacy architectures, eliminating bottlenecks using advanced profiling tools, and building cloud-native services that support millions of transactions daily. Experienced working across agile teams, mentoring junior engineers, and collaborating closely with DevOps, QA, and product teams to deliver robust and scalable backend platforms.

Skills

- Programming Languages: Rust, C#, Golang, Python, Java, VB.NET
- Frameworks: ASP.NET Core, Entity Framework, ReactJS, WebForms, Razor, .NET Framework, gRPC, Kafka, GraphQL
- Cloud & DevOps: AWS, Azure, Google Cloud Platform (GCP), Kubernetes (EKS), Docker, Prometheus, Grafana, Terraform
- APIs & Communication: RESTful APIs, gRPC, WebSockets, Kinesis, Async/Await
- Database Technologies: SQL Server, PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Cassandra, Linq to SQL
- Optimization Techniques: Advanced Vector Extensions (AVX), dynamic method generation, reducing allocations, batch processing, hash-based lookups
- Tools & Platforms: Visual Studio, JetBrains Rider, Git, Jenkins, SonarQube, Jira, Confluence
- Data Processing: Apache Kafka, Kinesis, ETL pipelines
- Testing & Debugging: xUnit, NUnit, Selenium, Postman, Fiddler, profiling tools (DotTrace, PerfView)
- Version Control & CI/CD: Git, GitHub, GitLab, Bitbucket, Jenkins, CircleCI

Experience

BACKEND TECHNICAL LEAD | QUANATA | NOVEMBER 2022 – PRESENT

- Spearheaded the architectural overhaul of legacy monolith systems into event-driven microservices using C#, Java, and Rust, hosted on Kubernetes clusters in AWS, improving system scalability and reducing deployment friction.
- Designed and implemented high-performance RESTful APIs and GraphQL endpoints capable of handling tens of thousands of concurrent requests with sub-100ms latency.

- Coordinated cross-functional backend squads (7 engineers) with clear ownership domains; led sprint planning, technical design reviews, and system reliability checks.
- Created reusable service templates in C# and Java to bootstrap new microservices with built-in logging, tracing, and health checks using Prometheus/Grafana stacks.
- Mentored junior developers and introduced internal workshops on optimizing .NET memory usage, async programming, and AWS-specific performance patterns.
- Drove adoption of feature flagging, chaos testing, and blue/green deployments to enable safer releases and experimentation.

Technologies Used: C#, Java, Rust, AWS (ECS, Lambda, Kinesis, RDS), GraphQL, REST APIs, Kubernetes, Docker, Prometheus, Grafana, GitHub Actions

SENIOR SOFTWARE ENGINEER | GRO INTELLIGENCE | APRIL 2022 – NOVEMBER 2022

- Led backend architecture for a data ingestion and analytics platform processing hundreds of terabytes of agriculture and climate data.
- Built batch and streaming data pipelines in Java and Rust integrated with Kafka and AWS Kinesis, applying filtering and transformation logic for ingestion.
- Designed schema validation services for incoming data streams using Spring Boot and implemented API gateways for data exposure to clients.
- Tuned PostgreSQL query execution plans, introduced read replicas, and optimized materialized views to support low-latency reporting.
- Contributed to migration from self-hosted infrastructure to managed Kubernetes clusters on EKS, enabling higher uptime and automated scaling.

Technologies Used: Java, Rust, Spring Boot, PostgreSQL, Kafka, AWS (Kinesis, RDS, EKS), Terraform, Kubernetes

PRINCIPAL SOFTWARE ENGINEER, OPTIMIZATION | NICE CXONE | JUNE 2021 – APRIL 2022

- Developed distributed backend systems using C# and .NET 6 for high-throughput logistics and fleet management platforms.
- Re-architected legacy services to use asynchronous I/O patterns and batch processing queues via AWS Kinesis, reducing processing latency by 45%.
- Built reusable libraries in C# for dynamic .NET IL method generation, replacing reflection-heavy logic and reducing allocations across services.
- Created and maintained Prometheus/Grafana dashboards visualizing job runtimes, queue depths, and memory usage.
- Designed and deployed gRPC-based interservice communication layer for high-efficiency service orchestration.
- Co-led a 5-member backend pod, conducting bi-weekly design reviews and performance regression audits.

Technologies Used: C#, .NET 6, Kinesis, Redis, gRPC, AWS, Prometheus, Grafana, Docker

FULL STACK ENGINEER | CONNECT FINANCIAL SOFTWARE SOLUTIONS SEPTEMBER 2018 – OCTOBER 2019

- Developed enterprise financial systems using C#, VB.NET, and ASP.NET MVC, including a custom rules engine and advanced form validation workflows.
- Designed and documented REST APIs for bank-to-core integrations and client dashboards.

- Maintained and refactored legacy Razor Pages and WebForms components into modern ReactJS interfaces, enabling faster feature delivery.
- Collaborated with QA, DevOps, and product teams using Agile/Scrum methodology, including TDD and weekly sprint retrospectives.

Technologies Used: C#, ASP.NET MVC, VB.NET, ReactJS, Linq to SQL, SQL Server, Jenkins

SOFTWARE ENGINEER | PLEXUS CORP. | MAY 2011 – DECEMBER 2015

- Worked closely with hardware engineering teams to write and test embedded system logic and device firmware in C and C++.
- Automated hardware diagnostics using Python scripts to simulate input loads and verify sensor outputs.
- Developed C# test harnesses and SQL Server diagnostic data loggers for embedded devices under QA testing.

Technologies Used: C, C++, C#, Python, SQL Server, Embedded Systems

Education

BACHELOR OF SCIENCE | 2007-2011 | VIRGINIA TECH

B.S. in Computer Science